BIM Command Center & AI Copilot

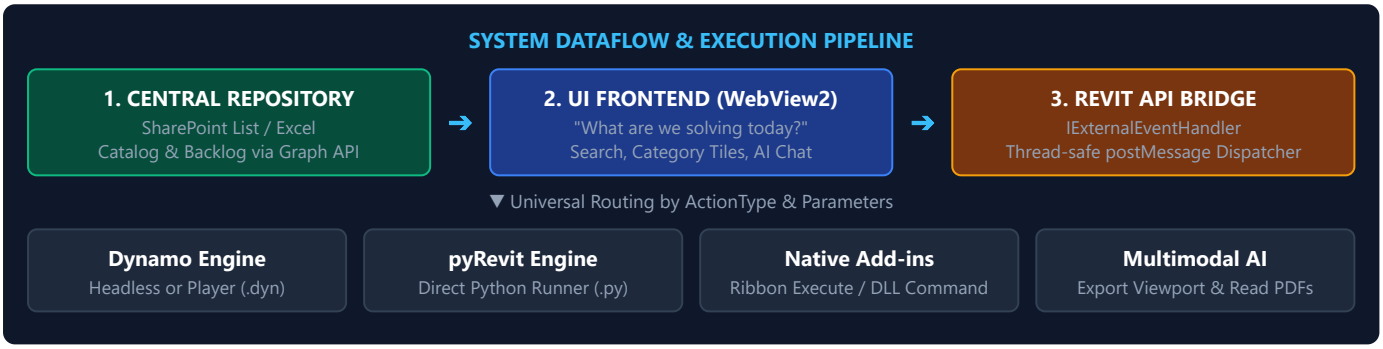
Universal Dispatcher, SharePoint Catalog Schema & Multimodal AI Architecture

SYSTEM ARCHITECTURE SPECIFICATION
Autodesk Revit (2022 - 2026+)
Status: Production Blueprint | v1.0.0

1 Executive Vision: "BIM App Store + Copilot"

Traditional deployment of computational design tools suffers from heavy friction: scripts are scattered across network drives, naming conventions are cryptic, Dynamo Player loads slowly, and distributing UI changes requires re-installing add-ins on every workstation.

> **Core Architectural Principle: Decoupled Thin-Client with Central Cloud Brain**
>
> A **single permanent Revit wrapper** hosts an embedded **Edge WebView2 modern web application**. All tool metadata (categories, descriptions, icons, paths, prerequisites) lives centrally in **SharePoint**. The interface serves as a **Universal Action Dispatcher** capable of running Dynamo scripts, pyRevit commands, third-party add-ins, native Revit commands, and an interactive multimodal AI Copilot.



2 Scalable Technology Stack Choices

Frontend Layer (UI/UX) — EDGE WEBVIEW2

- **Tech Stack:** React + Tailwind CSS + Lucide Icons (or Svelte/Vue).
- **Hosting:** Internal web server or Azure Static Web App.
- **Zero Recompile:** UI layout, search heuristics, category cards, and iconography update instantaneously across all users without restarting Revit.
- **Rich UX:** "What are we solving today?" intent bar, category tiles, instant fuzzy search (Fuse.js), and animated loading states.

Backend & Metadata Layer — SHAREPOINT LIST

- **Why List over Excel:** Native REST API (MS Graph), zero file-lock errors, granular permissions, built-in version control, and native Grid Edit UI for BIM managers.
- **Offline & Speed Strategy:** Client caches catalog locally in `catalog_cache.json` (TTL: 30 mins) with a manual "Refresh" button.
- **Automation Backlog:** A separate SharePoint List capturing user automation requests directly from the in-app idea button.

Execution Bridge (In-Session) — IEXTERNALEVENTHANDLER

- **Bridge Container:** pyRevit PushButton or compiled C# Ribbon Add-in.
- **IPC Protocol:** Bi-directional communication via WebView2's `window.chrome.webview.postMessage()`.
- **Thread Safety:** Strictly routes all Revit document modifications through an asynchronous `IExternalEventHandler` queue to prevent Revit API threading violations.

Multimodal AI Assistant — VISION + DOCUMENT AI

- **Active Viewport Capture:** Uses native `Document.ExportImage()` to capture high-res view screenshots into memory without third-party screen grab tools.
- **PDF Context:** Drag-and-drop architectural specs or codes into chat; text and raster extracted and passed to LLM context.
- **Function Calling:** LLM can suggest and run verified tools from the catalog.

3 Complete Database Schema Specification

The centralized SharePoint environment requires two lists: the **BIM Tool Catalog** and the **Automation Idea Backlog**.

Table 1: `BIM_ToolCatalog` (Primary Repository)

Field Name	Type	Req	Description & Constraints	Production Example
`ToolID`	Single Text	Yes	Unique slug identifier (kebab-case)	`doc-renumber-doors`
`Title`	Single Text	Yes	Human-readable display name	`Renumber Doors by Room`
`Category`	Choice	Yes	Enum: `Model Audit`, `Documentation`, `Geometry`, `QA/QC`, `Export`, `Setup`	`Documentation`
`Description`	Multi Text	Yes	Plain-language summary of what the tool accomplishes and prerequisites.	`Renumbers all door marks matching the adjacent room number and level prefix.`
`Keywords`	Multi Text	No	Comma-separated search keywords for fuzzy & AI matching.	`door, marks, renumber, room, schedule, audit`
`IconName`	Single Text	Yes	Lucide icon key or SVG URL.	`DoorClosed` or `https://cdn/.../door.svg`
`ActionType`	Choice	Yes	Enum: `Dynamo`, `PyRevit`, `NativeAddin`, `PostableCommand`, `ExternalLink`	`Dynamo`
`TargetPath`	Single Text	Yes	Relative path to script or Ribbon Command ID.	`\Documentation\RenumberDoors.dyn`
`ExecutionMode`	Choice	Yes	Enum: `Headless` (1-click silent run), `Player` (opens Dynamo Player for inputs)	`Headless`
`RevitVersions`	Multi Choice	Yes	Multi-select: `2022`, `2023`, `2024`, `2025`, `2026`	`2024; 2025; 2026`
`PackagesRequired`	Single Text	No	Dynamo packages required for validation.	`Clockwork (2.4.0), Rhythm (2.4.1)`
`Status`	Choice	Yes	Enum: `Active`, `Beta`, `UnderReview`, `Deprecated`	`Active`
`CanAITrigger`	Boolean	Yes	Whether the AI Chatbot is permitted to invoke this tool autonomously.	`True`
`Version`	Single Text	Yes	Semantic version string.	`v2.1.0`

Table 2: `BIM_IdeaBacklog` (Crowdsourced Automation Pipeline)

Field Name	Type	Description	Production Example
`IdeaTitle`	Single Text	Short summary of the requested automation.	`Batch Align Sheet Viewports`
`ProblemStatement`	Multi Text	Pain point, hours spent manually, and target outcome.	`Aligning plans across 40 sheets takes 3 hours per submission.`
`CategorySuggested`	Choice	Target category from catalog enum.	`Documentation`
`SubmitterEmail`	Single Text	Authenticated user email (auto-populated via SSO).	`user@company.com`
`RevitVersionUsed`	Single Text	Current Revit build of the user session.	`Revit 2025.2`
`AttachedScreenshot`	Attachments	Optional viewport capture or markup uploaded with idea.	`viewport_markup.png`
`Status`	Choice	Enum: `New`, `Evaluating`, `In Development`, `Completed`, `Rejected`	`Evaluating`

4 Bridge IPC Protocol & Multimodal Dispatcher

The communication between the Edge WebView2 frontend and Revit's C#/pyRevit bridge uses structured JSON messages over `postMessage`:

```
// 1. Tool Execution Payload (Frontend -> Revit Bridge)
{
  "action": "EXECUTE_TOOL",
  "toolId": "doc-renumber-doors",
  "actionType": "Dynamo",
  "targetPath": "C:\\BIM_Tools\\Documentation\\RenumberDoors.dyn",
  "executionMode": "Headless",
  "parameters": { "prefix": "D-", "level": "Level 1" }
}

// 2. Viewport Screen Capture Request (AI Chatbot -> Revit Bridge)
{
  "action": "CAPTURE_ACTIVE_VIEWPORT",
  "dpi": 150,
  "cropToVisibleRegion": true
}

// 3. Viewport Capture Response (Revit Bridge -> Frontend)
{
  "action": "VIEWPORT_CAPTURED",
  "mimeType": "image/png",
  "base64Data": "iVBORw0KGgoAAAANSUhEUgAA...",
  "activeViewName": "Level 2 - Architectural Plan"
}
```

5 Universal Dispatcher Implementation Reference

A. Triggering Native Add-in Buttons

Simulate a click on any installed Revit plugin button via Ribbon API:

```
using Autodesk.Windows;

public void ExecuteRibbonButton(string buttonId) {
    var ribbon = ComponentManager.Ribbon;
    var item = ribbon.FindItem(buttonId, true)
               as RibbonButton;
    if (item != null && item.IsEnabled) {
        item.Execute();
    }
}
```

B. Capturing Active Viewport for AI

Native Revit API capture without external screen grabbing utilities:

```
ImageExportOptions opts = new ImageExportOptions {
    ExportRange = ExportRange.VisibleRegionOfCurrentView,
    FilePath = tempPngPath,
    ImageResolution = ImageResolution.DPI_150,
    PixelSize = 1920
};
doc.ExportImage(opts);
byte[] bytes = File.ReadAllBytes(tempPngPath);
string base64 = Convert.ToBase64String(bytes);
```

6 Scalable Deployment & Rollout Strategy

Option A: pyRevit Extension Sync (Fastest)

Deploy a custom pyRevit extension pointing to a central read-only network folder:

`pyrevit extend ui CompanyTools \\server\BIM\pyRevit-Tools`

Pros: Immediate setup, zero MSI packaging, updates deploy to network folder and propagate instantly.

Option B: Lightweight C# .NET Add-in (Enterprise)

A single compiled DLL installed via Intune/SCCM once per workstation.

`C:\ProgramData\Autodesk\Revit\Addins\[Year]\BIMCommandCenter.addin`

Pros: 100% independent of pyRevit runtime, native enterprise management, highest execution speed.

Summary of Strategic Advantages

1. **Central Governance:** BIM managers modify scripts, update categories, and publish new tools in SharePoint without interrupting production staff.
2. **Zero Friction Tool Discovery:** Users search by intent ("What are we solving today?") rather than guessing file paths.
3. **Future-Proof Multimodal Intelligence:** The architecture supports immediate expansion into multimodal AI reasoning (viewport screenshots + PDF specifications) with zero structural rework.